Linear Algebra

Prithwijit Guha
Dept. of EEE, IIT Guwahati

About Data

- What is your Weight?
  - Answer is a Real Number (Scalar)
- What is the Date?
  - Represented by Three Numbers [*Date* , *Month* , *Year*]
  - Can be called n-Tuple (n=3)
  - The Order Matters!
- Temporally Ordered Audio Signals
- Data in Tables or Spreadsheets
- Monochrome or Color (Digital) Images
- Digital Video Data

Data Types & Indices

x, y, z : Scalar Data

$\{x_1, \ldots x_i, \ldots x_\mathrm{n}\}$: Indexed Set of Numbers

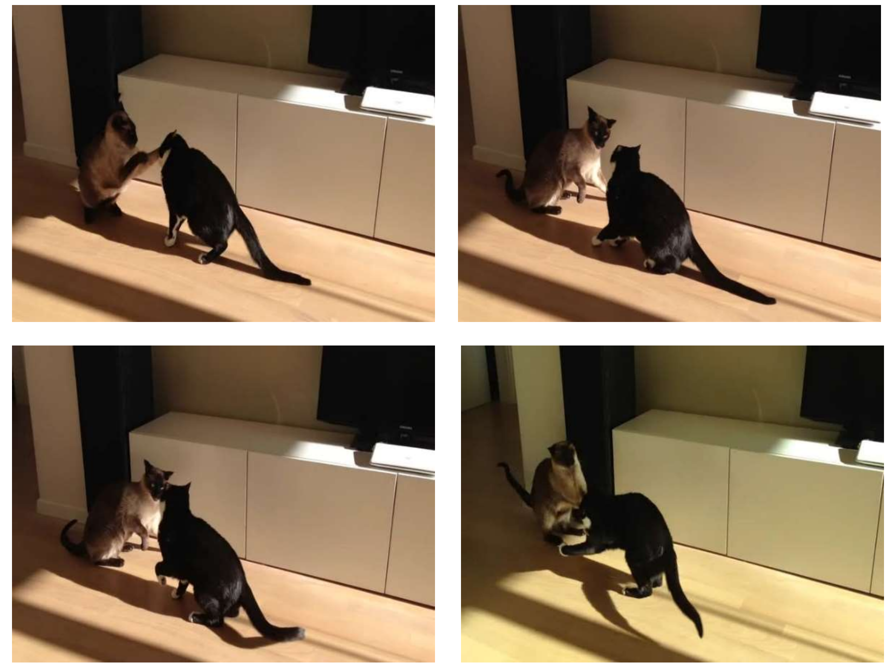
$\{x_1, x_2, x_3\} \neq \{x_2, x_1, x_3\}$: Order Matters

$(1,3,2018) \neq (3,1,2018)$: Date

Data Types & Indices

$\{x_t; t = 1, \dots\}$
Temporally Indexed
Scalar Data

Image as Spatially
Indexed (x, y) array
of Pixels



Video or Frame Sequence as
Spatio-Temporally Indexed
(x, y, t) Data

Vector Space

A Real/Complex Vector Space is a Set $\boldsymbol{V}$ with a Special Element $\mathbf{0}$

- Addition: IF $\boldsymbol{x}, \boldsymbol{y} \in \boldsymbol{V}$, THEN $\boldsymbol{x} + \boldsymbol{y} \in \boldsymbol{V}$
- Inverse: IF $\boldsymbol{x} \in \boldsymbol{V}$, THEN $\exists \boldsymbol{y} \in \boldsymbol{V}$ SUCH THAT $\boldsymbol{x} + \boldsymbol{y} = \mathbf{0}$
- Scalar Multiplication: IF $\boldsymbol{x} \in \boldsymbol{V}$, THEN $\mathrm{c}\boldsymbol{x} \in \boldsymbol{V}$ where $\boldsymbol{c}$ is Scalar

Additive Axioms

For every $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \boldsymbol{V}$

- $\boldsymbol{x} + \boldsymbol{y} = \boldsymbol{y} + \boldsymbol{x}$ **(Commutative Property)**
- $(\boldsymbol{x} + \boldsymbol{y}) + \boldsymbol{z} = \boldsymbol{x} + (\boldsymbol{y} + \boldsymbol{z})$ **(Associative Property)**
- $\mathbf{0} + \boldsymbol{x} = \boldsymbol{x} + \mathbf{0} = \boldsymbol{x}$ **(Identity)**
- $(-\boldsymbol{x}) + \boldsymbol{x} = \boldsymbol{x} + (-\boldsymbol{x}) = \mathbf{0}$ **(Inverse)**

Multiplicative Axioms

For every $\boldsymbol{x} \in \boldsymbol{V}$ and Real/Complex Number $\boldsymbol{c}, \boldsymbol{d}$

- $\mathbf{0} \cdot \boldsymbol{x} = \mathbf{0}$
- $\mathbf{1} \cdot \boldsymbol{x} = \boldsymbol{x}$
- $\boldsymbol{c}(\boldsymbol{d}\boldsymbol{x}) = \boldsymbol{c}\boldsymbol{d}(\boldsymbol{x}) = \boldsymbol{d}(\boldsymbol{c}\boldsymbol{x}) = \boldsymbol{c}\boldsymbol{d}\boldsymbol{x}$

Distributive Axioms

For every $\boldsymbol{x} \in \boldsymbol{V}$ and Real/Complex Number $\boldsymbol{c}, \boldsymbol{d}$

- $\boldsymbol{c}(\boldsymbol{x}+\boldsymbol{y}) = \boldsymbol{cx}+\boldsymbol{cy}$
- $(\boldsymbol{c}+\boldsymbol{d})\boldsymbol{x} = \boldsymbol{cx}+\boldsymbol{dx}$

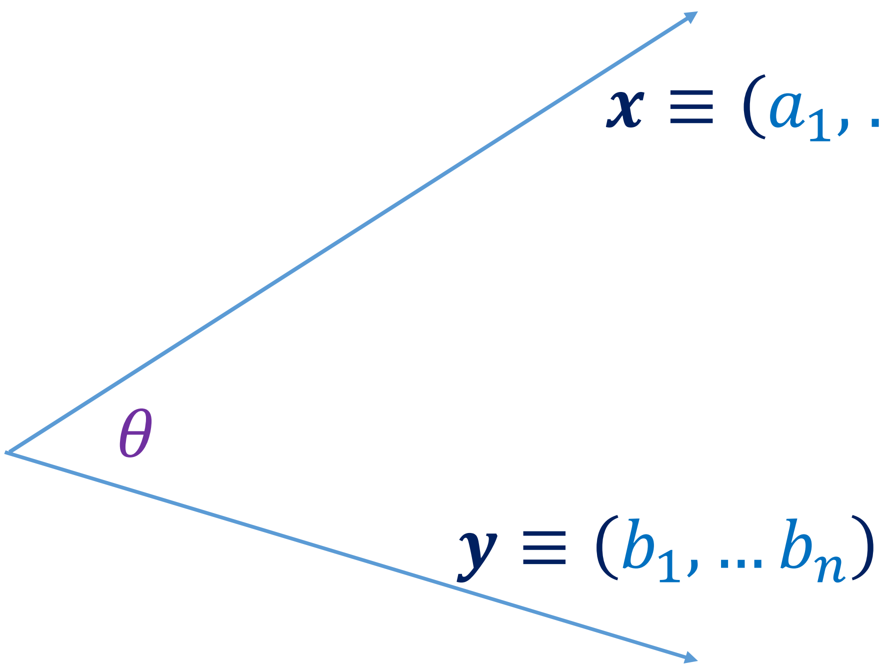
Norm of a Vector

Alternatively Length or Magnitude of a Vector

l^p Norm $\quad \|\boldsymbol{x}\|_p = \left(\sum_{r=1}^{n} |x[r]|^p\right)^{\frac{1}{p}}$

$$\boldsymbol{x} = \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix} \Rightarrow \|\boldsymbol{x}\|_2 = \sqrt{|1|^2 + |-2|^2 + |4|^2} = 4.583$$

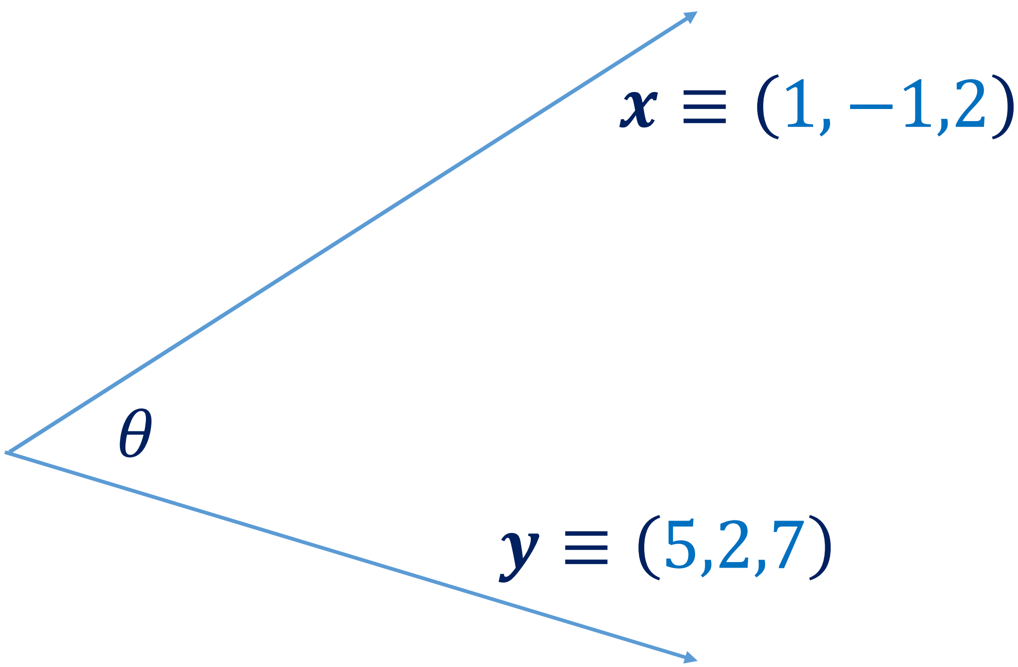
Inner Product



$$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \|\boldsymbol{x}\|_2 \|\boldsymbol{y}\|_2 \cos \boldsymbol{\theta}$$

$$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \sum_{i=1}^{n} a_i b_i$$

Inner Product

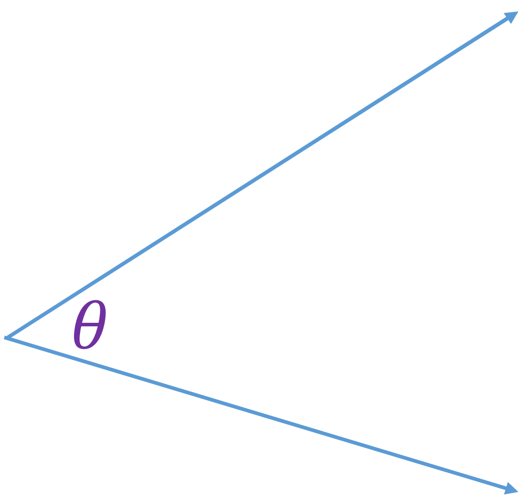


$$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = ?$$

$$\theta = ?$$

Inner Product

$$\mathbf{x} \equiv (1, -1, 2)$$



$$\mathbf{y} \equiv (5, 2, 7)$$

$$\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=1}^{n} x_i y_i = 17$$

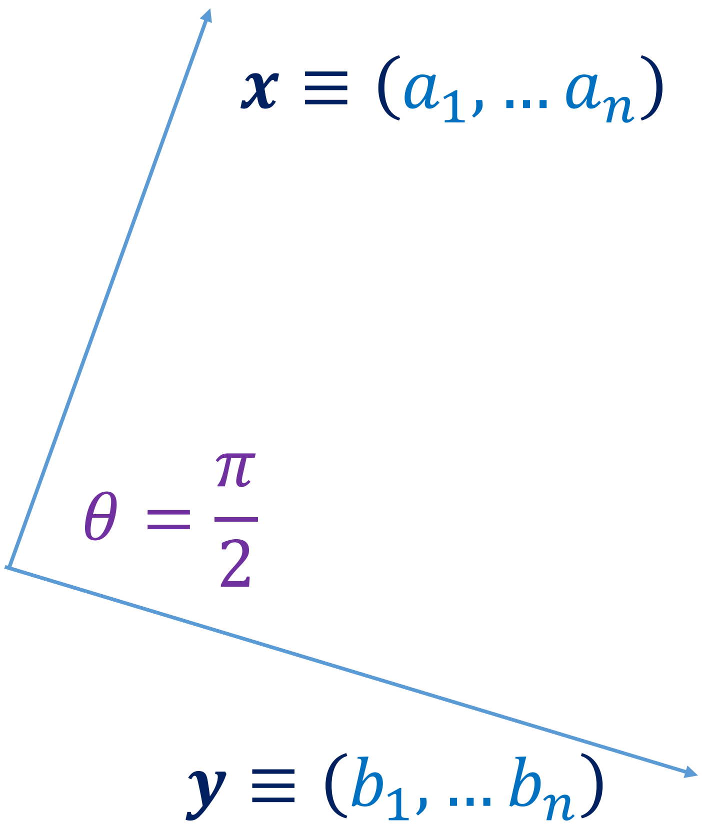
$$\langle \mathbf{x}, \mathbf{y} \rangle = \|\mathbf{x}\|_2 \|\mathbf{y}\|_2 \cos \boldsymbol{\theta}$$

$$\cos \boldsymbol{\theta} = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{x}\|_2 \|\mathbf{y}\|_2}$$

$$\theta = cos^{-1}\left(\frac{17}{2.45 \times 8.83}\right)$$

$$\theta = 38^{o}$$

Orthogonal Vectors

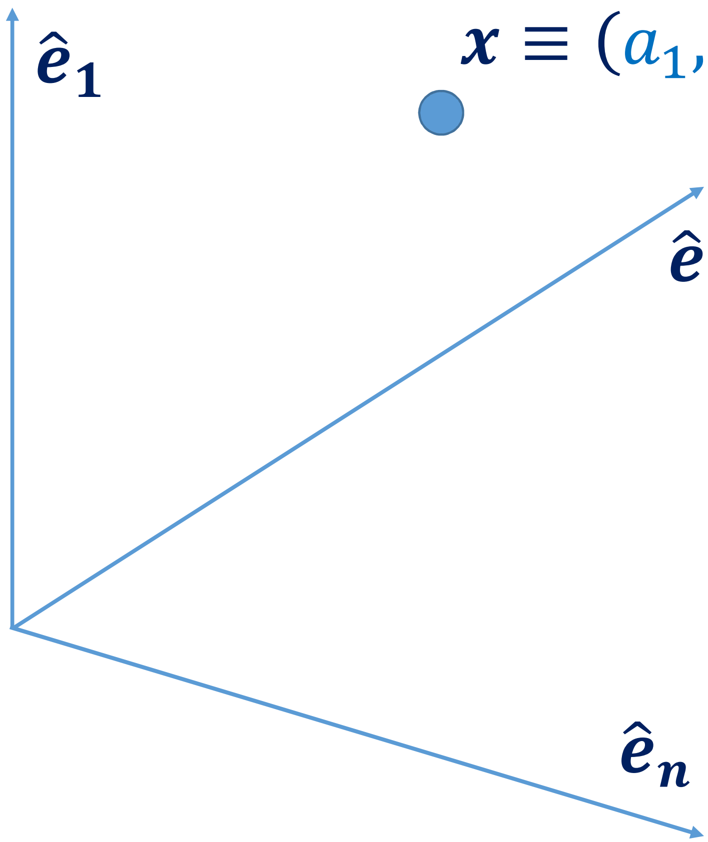


$$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \sum_{i=1}^{n} a_i b_i = 0$$

$$\langle \boldsymbol{x}, \boldsymbol{y} \rangle = \|\boldsymbol{x}\|_2 \|\boldsymbol{y}\|_2 \cos \frac{\pi}{2} = 0$$

Basis Vectors

$$\boldsymbol{x} \equiv (a_1, a_2, \dots a_n)$$

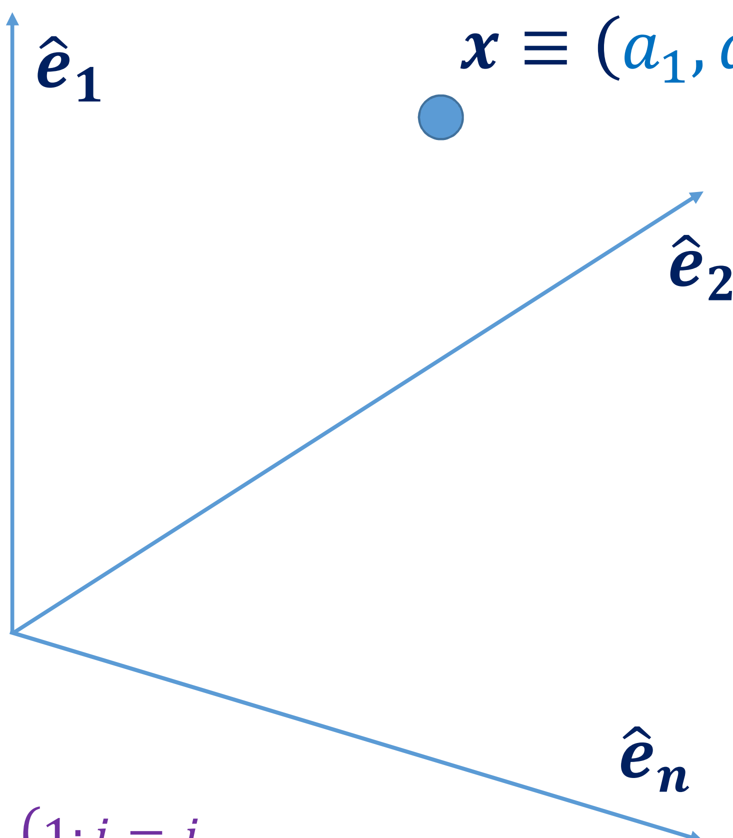


$$\|\hat{\boldsymbol{e}}_{\boldsymbol{i}}\|_2 = 1$$

$$\boldsymbol{x} = a_1\hat{\boldsymbol{e}}_{\boldsymbol{1}} + a_2\hat{\boldsymbol{e}}_{\boldsymbol{2}} + \cdots + a_n\hat{\boldsymbol{e}}_{\boldsymbol{n}}$$

Orthogonal Basis Vectors

$$\boldsymbol{x} \equiv (a_1, a_2, \ldots a_n)$$



$$\|\hat{\boldsymbol{e}}_{\boldsymbol{i}}\|_2 = 1$$

$$\langle \hat{\boldsymbol{e}}_{\boldsymbol{i}}, \hat{\boldsymbol{e}}_{\boldsymbol{j}} \rangle = \delta[i - j]$$

$$a_i = \langle \boldsymbol{x}, \hat{\boldsymbol{e}}_{\boldsymbol{i}} \rangle$$

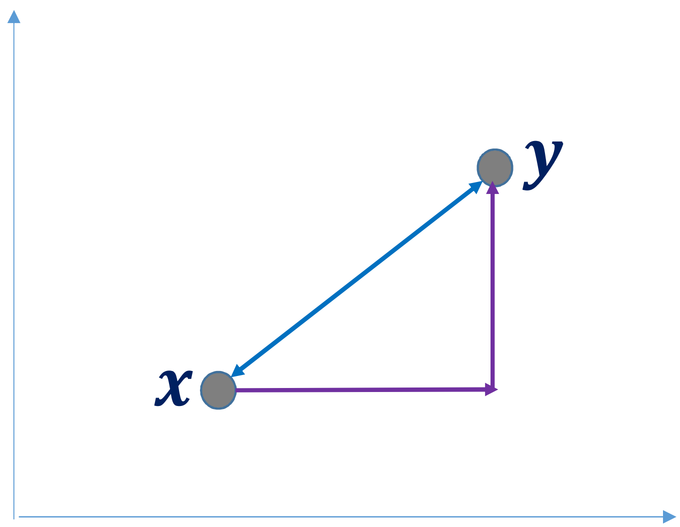
$$\boldsymbol{x} = a_1 \hat{\boldsymbol{e}}_{\mathbf{1}} + a_2 \hat{\boldsymbol{e}}_{\mathbf{2}} + \cdots + a_n \hat{\boldsymbol{e}}_{\boldsymbol{n}}$$

$$\delta[i - j] = \begin{cases} 1; i = j \\ 0; i \neq j \end{cases}$$

Distance between Two Vectors

- Required for Classification Problems
- Metric on $\boldsymbol{V}$ is a Function $\boldsymbol{d}: \boldsymbol{V} \times \boldsymbol{V} \rightarrow [\mathbf{0}, \infty)$
- Function $d(\boldsymbol{x}, \boldsymbol{y})$ satisfies the following for $\boldsymbol{x}, \boldsymbol{y} \in \boldsymbol{V}$
  - $\boldsymbol{d}(\boldsymbol{x}, \boldsymbol{y}) \geq 0$
  - $\boldsymbol{d}(\boldsymbol{x}, \boldsymbol{y}) = \mathbf{0} \iff \boldsymbol{x} = \boldsymbol{y}$
  - $\boldsymbol{d}(\boldsymbol{x}, \boldsymbol{y}) = \boldsymbol{d}(\boldsymbol{y}, \boldsymbol{x})$
  - $\boldsymbol{d}(\boldsymbol{x}, \boldsymbol{y}) \leq \boldsymbol{d}(\boldsymbol{x}, \boldsymbol{z}) + \boldsymbol{d}(\boldsymbol{y}, \boldsymbol{z})$

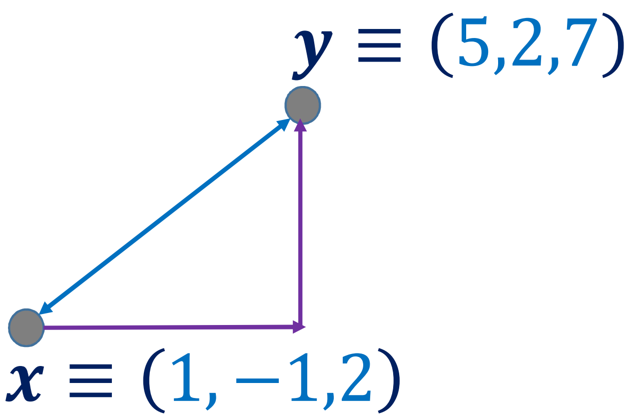
Euclidean and Manhattan Distance



$$d_E^2(\boldsymbol{x}, \boldsymbol{y}) = \sum_{i=1}^{n} (\boldsymbol{x}_i - \boldsymbol{y}_i)^2$$

$$d_M(\boldsymbol{x}, \boldsymbol{y}) = \sum_{i=1}^{n} |\boldsymbol{x}_i - \boldsymbol{y}_i|$$

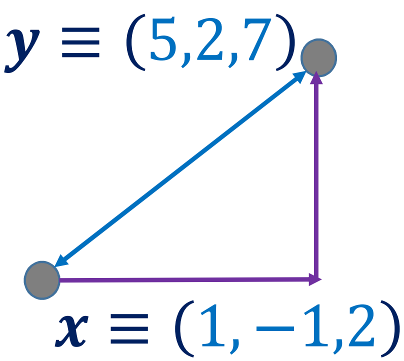
Euclidean and Manhattan Distance



$$d_E(\boldsymbol{x}, \boldsymbol{y}) = ?$$

$$d_M(\boldsymbol{x}, \boldsymbol{y}) = ?$$

Euclidean and Manhattan Distance



$$d_E^2(\boldsymbol{x},\boldsymbol{y}) = (1-5)^2 + (-1-2)^2 + (2-7)^2$$

$$d_E^2(\boldsymbol{x},\boldsymbol{y}) = 16 + 9 + 25 = 50$$

$$d_E(\boldsymbol{x},\boldsymbol{y}) = 7.07$$

$$d_M(\boldsymbol{x},\boldsymbol{y}) = |1-5| + |-1-2| + |2-7|$$

$$d_M(\boldsymbol{x},\boldsymbol{y}) = 4 + 3 + 5$$

$$d_M(\boldsymbol{x},\boldsymbol{y}) = 12$$

Weighted Euclidean Distance

In Practical Applications, we want to Scale the Dimensions

$$d^2(\boldsymbol{x}, \boldsymbol{y}) = \sum_{i=1}^{n} (\boldsymbol{x}_i - \boldsymbol{y}_i)^2$$

$$d^2(\boldsymbol{x}, \boldsymbol{y}) = \sum_{i=1}^{n} s_i (\boldsymbol{x}_i - \boldsymbol{y}_i)^2$$

Matrix

A Matrix is a Two Dimensional Arrangement of Numbers

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1j} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{i1} & \cdots & a_{ij} & \cdots & a_{in} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mj} & \cdots & a_{mn} \end{bmatrix}$$

Square Matrix

$$m = n$$

Row Vector

$$A(i, :) = [a_{i1} \quad \cdots \quad a_{in}]$$

Column Vector

$$A(:, j) = \begin{bmatrix} a_{1j} \\ \vdots \\ a_{mj} \end{bmatrix}$$

Matrix Transpose

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

$$A^T = \begin{bmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{bmatrix}$$

Transpose of Matrix A

Symmetric Matrix: $A = A^T$

Example: $A = \begin{bmatrix} 1 & 2 & 3 \\ 11 & 12 & 13 \end{bmatrix} \Rightarrow A^T = \begin{bmatrix} 1 & 11 \\ 2 & 12 \\ 3 & 13 \end{bmatrix}$

Matrix Transpose

$$A = \begin{bmatrix} 4 & -8 \\ -3 & 0 \\ 11 & -5 \end{bmatrix} \qquad A^T = ?$$

Matrix Transpose

$$A = \begin{bmatrix} 4 & -8 \\ -3 & 0 \\ 11 & -5 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 4 & -3 & 11 \\ -8 & 0 & -5 \end{bmatrix}$$

Matrix Transpose

How will you Generate a Random Symmetric Matrix?

Symmetric Matrix: $A = A^T$

Matrix Transpose

Generate Random Matrix: A

$$A = \begin{bmatrix} -1 & 3 & 0 \\ 2 & 5 & -7 \\ -9 & 10 & 4 \end{bmatrix}$$

Evaluate: A^T

$$A^T = \begin{bmatrix} -1 & 2 & -9 \\ 3 & 5 & 10 \\ 0 & -7 & 4 \end{bmatrix}$$

Compute: $\mathrm{B} = \mathrm{A} + A^T$

$$B = \begin{bmatrix} -2 & 5 & -9 \\ 5 & 10 & 3 \\ -9 & 3 & 8 \end{bmatrix}$$

Check: $\mathrm{B} = B^T$

Matrix Operations: Scalar Multiplication

Multiplication of Matrix A and a Scalar $c \in \mathbb{R}^1$

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

$$c \cdot A = A \cdot c = \begin{bmatrix} ca_{11} & ca_{12} & \cdots & ca_{1n} \\ ca_{21} & ca_{22} & \cdots & ca_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ ca_{m1} & ca_{m2} & \cdots & ca_{mn} \end{bmatrix}$$

Matrix Operations: Addition

Matrices A and B are of the same size $m \times n$, and $c, d \in \mathbb{R}^1$

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \qquad B = \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{m1} & b_{m2} & \cdots & b_{mn} \end{bmatrix}$$

$$c \cdot A + d \cdot B = \begin{bmatrix} ca_{11} + db_{11} & ca_{12} + db_{12} & \cdots & ca_{1n} + db_{1n} \\ ca_{21} + db_{21} & ca_{22} + db_{22} & \cdots & ca_{2n} + db_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ ca_{m1} + db_{m1} & ca_{m2} + db_{m2} & \cdots & ca_{mn} + db_{mn} \end{bmatrix}$$

Matrix Operations: Addition

$$A = \begin{bmatrix} 1 & 5 & 2 \\ -2 & 3 & 7 \end{bmatrix} \qquad B = \begin{bmatrix} 2 & -3 & 7 \\ -1 & -2 & 6 \end{bmatrix}$$

Evaluate $\mathrm{C} = 2A - 3B$

Matrix Operations: Addition

$$C = 2 * A + (-3) * B$$

$$C = (2) * \begin{bmatrix} 1 & 5 & 2 \\ -2 & 3 & 7 \end{bmatrix} + (-3) * \begin{bmatrix} 2 & -3 & 7 \\ -1 & -2 & 6 \end{bmatrix}$$

$$C = \begin{bmatrix} 2 & 10 & 4 \\ -4 & 6 & 14 \end{bmatrix} + \begin{bmatrix} -6 & 9 & -21 \\ 3 & 6 & -18 \end{bmatrix}$$

$$C = \begin{bmatrix} -4 & 19 & -17 \\ -1 & 12 & -4 \end{bmatrix}$$

Matrix as Linear Operator

$A \in \mathbb{R}^{m\times n}$

$y \in \mathbb{R}^m \longrightarrow$ $$y = Ax$$

$x \in \mathbb{R}^n$

$$A\colon \mathbb{R}^n \to \mathbb{R}^m$$

$$\boldsymbol{y} = A(\alpha \boldsymbol{x}_1 + \beta \boldsymbol{x}_2) = \alpha(A\boldsymbol{x}_1) + \beta(A\boldsymbol{x}_2)$$

Matrix Operations: Matrix-Vector Multiplication

Matrix A of size $m \times n$ and Column Vector $\boldsymbol{x}$ of size $n \times 1$

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \qquad \boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \qquad \boldsymbol{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix} = A\boldsymbol{x}$$

$$y_i = \sum_{j=1}^{n} a_{ij} \times x_j$$

Matrix Operations: Matrix-Vector Multiplication

$$A = \begin{bmatrix} 1 & 2 & -3 \\ 7 & -1 & 2 \end{bmatrix} \qquad \boldsymbol{x} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$$

$$\boldsymbol{y} = A\boldsymbol{x} = \begin{bmatrix} 1 \times 1 + 2 \times (-1) + (-3) \times 2 \\ 7 \times 1 + (-1) \times (-1) + 2 \times 2 \end{bmatrix} = \begin{bmatrix} -7 \\ 12 \end{bmatrix}$$

Matrix Operations: Multiplication

Matrices A and B are of respective sizes $m \times n$ and $n \times p$

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \qquad B = \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1p} \\ b_{21} & b_{22} & \cdots & b_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{np} \end{bmatrix}$$

Matrix $\mathrm{C} = A \times B$ is of size $m \times p$

$$c_{ij} = \sum_{k=1}^{n} a_{ik} \times b_{kj}$$

Matrix Operations: Multiplication

$$A = \begin{bmatrix} 1 & 2 & -3 \\ 7 & -1 & 2 \end{bmatrix}$$

$$B = \begin{bmatrix} -2 & 5 \\ 2 & -1 \\ 1 & 3 \end{bmatrix}$$

$$C = A * B = ?$$

Matrix Operations: Multiplication

$$A = \begin{bmatrix} 1 & 2 & -3 \\ 7 & -1 & 2 \end{bmatrix} \qquad B = \begin{bmatrix} -2 & 5 \\ 2 & -1 \\ 1 & 3 \end{bmatrix}$$

$$C = A * B$$

$$C = \begin{bmatrix} 1 \times (-2) + 2 \times 2 + (-3) \times 1 & 1 \times 5 + 2 \times (-1) + (-3) \times 3 \\ 7 \times (-2) + (-1) \times 2 + 2 \times 1 & 7 \times 5 + (-1) \times (-1) + 2 \times 3 \end{bmatrix}$$

$$C = \begin{bmatrix} -1 & -6 \\ -14 & 42 \end{bmatrix}$$

Matrix Operations: Multiplication

$$\boldsymbol{x} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} \qquad \boldsymbol{y} = \begin{bmatrix} 5 \\ 2 \\ 7 \end{bmatrix}$$

$$\boldsymbol{x}^T\boldsymbol{y} = ? \qquad \boldsymbol{x}\boldsymbol{y}^T = ?$$

Matrix Operations: Multiplication

$$\boldsymbol{x}^T\boldsymbol{y} = \langle \boldsymbol{x}, \boldsymbol{y} \rangle = \begin{bmatrix} 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 5 \\ 2 \\ 7 \end{bmatrix} = 17$$

$$\boldsymbol{x}\boldsymbol{y}^T = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} \begin{bmatrix} 5 & 2 & 7 \end{bmatrix} = \begin{bmatrix} 5 & 2 & 7 \\ -5 & -2 & -7 \\ 10 & 4 & 14 \end{bmatrix}$$

Matrix Operations: Multiplication

In General $\mathrm{AB} \neq BA$, might be possible in special cases

Associativity: $\mathrm{A(BC)} = (AB)C$

Distributivity: $\mathrm{A(B + C)} = AB + AC$ and $\mathrm{(A + B)}C = AC + BC$

Transpose of Product: $(AB)^T = B^T A^T$ and $\left(\prod_{i=1}^{n} A_i\right)^T = \prod_{j=0}^{n-1} A_{n-j}^T$

Multiplication Rules Must be Strictly Followed

Linear Independence

Vectors $\boldsymbol{v}_1, \dots \boldsymbol{v}_i, \dots \boldsymbol{v}_n$ are Linearly Independent

$$\Downarrow$$

$$\sum_{i=1}^{n} \alpha_i \boldsymbol{v}_i = 0 \quad \Rightarrow \quad \forall i \ \alpha_i = 0$$

Matrix Operations: Determinant

$$B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\det(B) = |B| = ad - bc$$

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

Minors

$$\det(A) = a \times det\left(\begin{bmatrix} e & f \\ h & i \end{bmatrix}\right) - b \times det\left(\begin{bmatrix} d & f \\ g & i \end{bmatrix}\right) + c \times det\left(\begin{bmatrix} d & e \\ g & h \end{bmatrix}\right)$$

Minor Expansion Formula

Matrix Operations: Determinant

$$A = \begin{bmatrix} 1 & -2 & 5 \\ 7 & -3 & 2 \\ -1 & 2 & 0 \end{bmatrix} \qquad |A| = ?$$

Matrix Operations: Determinant

$$A = \begin{bmatrix} 1 & -2 & 5 \\ 7 & -3 & 2 \\ -1 & 2 & 0 \end{bmatrix}$$

$$|A| = 1 \times \begin{bmatrix} -3 & 2 \\ 2 & 0 \end{bmatrix} - (-2) \times \begin{bmatrix} 7 & 2 \\ -1 & 0 \end{bmatrix} + 5 \times \begin{bmatrix} 7 & -3 \\ -1 & 2 \end{bmatrix}$$

$$|A| = 1 \times \{(-3 \times 0) - (2 \times 2)\} + 2 \times \{(7 \times 0) - (-1 \times 2)\} + 5 \times \{(7 \times 2) - (-1 \times -3)\}$$

$$|A| = -4 + 4 + 55 = 55$$

Matrix Operations: Determinant

$$A = \begin{bmatrix} 5 & 2 & 7 \\ -5 & -2 & -7 \\ 10 & 4 & 14 \end{bmatrix} \qquad |A| = ?$$

Matrix Operations: Inverse

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\boldsymbol{C} = \begin{bmatrix} d & -c \\ -b & a \end{bmatrix}$$

Cofactor Matrix of A

Minor of A_{ij}

$$C_{ij} = (-1)^{i+j}|M_{ij}|$$

$$adj(A) = \boldsymbol{C}^T = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Matrix Operations: Inverse

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \qquad adj(A) = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$A \times adj(A) = \begin{bmatrix} ad - bc & 0 \\ 0 & ad - bc \end{bmatrix} = |A| \times I_2$$

$$A^{-1} \times A \times adj(A) = |A| \times I_2 \times A^{-1}$$

$$A^{-1} = \frac{adj(A)}{|A|}$$

Matrix Operations: Inverse

$$A = \begin{bmatrix} 1 & -2 & 5 \\ 7 & -3 & 2 \\ -1 & 2 & 0 \end{bmatrix} \qquad A^{-1} = ?$$

Matrix Operations: Inverse – Cofactor Matrices

$$M_{11} = \begin{bmatrix} -3 & 2 \\ 2 & 0 \end{bmatrix} \qquad M_{12} = \begin{bmatrix} 7 & 2 \\ -1 & 0 \end{bmatrix} \qquad M_{13} = \begin{bmatrix} 7 & -3 \\ -1 & 2 \end{bmatrix}$$

$$M_{21} = \begin{bmatrix} -2 & 5 \\ 2 & 0 \end{bmatrix} \qquad M_{22} = \begin{bmatrix} 1 & 5 \\ -1 & 0 \end{bmatrix} \qquad M_{23} = \begin{bmatrix} 1 & -2 \\ -1 & 2 \end{bmatrix}$$

$$M_{31} = \begin{bmatrix} -2 & 5 \\ -3 & 2 \end{bmatrix} \qquad M_{32} = \begin{bmatrix} 1 & 5 \\ 7 & 2 \end{bmatrix} \qquad M_{33} = \begin{bmatrix} 1 & -2 \\ 7 & -3 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & -2 & 5 \\ 7 & -3 & 2 \\ -1 & 2 & 0 \end{bmatrix}$$

Matrix Operations: Inverse – Cofactor Matrices

$$M_{11} = \begin{bmatrix} -3 & 2 \\ 2 & 0 \end{bmatrix} = -4 \quad M_{12} = \begin{bmatrix} 7 & 2 \\ -1 & 0 \end{bmatrix} = 2 \quad M_{13} = \begin{bmatrix} 7 & -3 \\ -1 & 2 \end{bmatrix} = 11$$

$$M_{21} = \begin{bmatrix} -2 & 5 \\ 2 & 0 \end{bmatrix} = -10 \quad M_{22} = \begin{bmatrix} 1 & 5 \\ -1 & 0 \end{bmatrix} = 5 \quad M_{23} = \begin{bmatrix} 1 & -2 \\ -1 & 2 \end{bmatrix} = 0$$

$$M_{31} = \begin{bmatrix} -2 & 5 \\ -3 & 2 \end{bmatrix} = 11 \quad M_{32} = \begin{bmatrix} 1 & 5 \\ 7 & 2 \end{bmatrix} = -33 \quad M_{33} = \begin{bmatrix} 1 & -2 \\ 7 & -3 \end{bmatrix} = 11$$

$$A = \begin{bmatrix} 1 & -2 & 5 \\ 7 & -3 & 2 \\ -1 & 2 & 0 \end{bmatrix} \qquad C_{ij} = (-1)^{i+j}\left|M_{ij}\right| \qquad C = \begin{bmatrix} -4 & -2 & 11 \\ 10 & 5 & 0 \\ 11 & 33 & 11 \end{bmatrix}$$

Matrix Operations: Inverse – Adjoint Matrix

$$C = \begin{bmatrix} -4 & -2 & 11 \\ 10 & 5 & 0 \\ 11 & 33 & 11 \end{bmatrix}$$

$$adj(A) = \boldsymbol{C}^T$$

$$adj(A) = \begin{bmatrix} -4 & 10 & 11 \\ -2 & 5 & 33 \\ 11 & 0 & 11 \end{bmatrix}$$

Matrix Operations: Inverse

$$adj(A) = \begin{bmatrix} -4 & 10 & 11 \\ -2 & 5 & 33 \\ 11 & 0 & 11 \end{bmatrix}$$

$$A^{-1} = \frac{adj(A)}{|A|}$$

$$A^{-1} = \frac{1}{55}\begin{bmatrix} -4 & 10 & 11 \\ -2 & 5 & 33 \\ 11 & 0 & 11 \end{bmatrix} = \begin{bmatrix} -0.073 & 0.182 & 0.2 \\ -0.036 & 0.091 & 0.6 \\ 0.2 & 0 & 0.2 \end{bmatrix}$$

Equation Solving

$$a_{11}x_1 + a_{12}x_2 + \cdots a_{1n}x_n = b_1$$
$$a_{21}x_1 + a_{22}x_2 + \cdots a_{2n}x_n = b_2$$

n-Variables , n-Equations

$$a_{n1}x_1 + a_{n2}x_2 + \cdots a_{nn}x_n = b_n$$

$$Ax = b$$

$$x = A^{-1}b$$

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \qquad b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

Equation Solving

$$x - 2y + 5z = 2$$

$$7x - 3y + 2z = 5$$

$$2y - x = 3$$

$$x, y, z = ?$$

Equation Solving

$$x - 2y + 5z = 2$$

$$7x - 3y + 2z = 5$$

$$2y - x = 3$$

$$\underbrace{\begin{bmatrix} 1 & -2 & 5 \\ 7 & -3 & 2 \\ -1 & 2 & 0 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x \\ y \\ z \end{bmatrix}}_{P} = \underbrace{\begin{bmatrix} 2 \\ 5 \\ 3 \end{bmatrix}}_{B}$$

$$AP = B \Rightarrow P = A^{-1}B$$

$$P = \begin{bmatrix} -0.073 & 0.182 & 0.2 \\ -0.036 & 0.091 & 0.6 \\ 0.2 & 0 & 0.2 \end{bmatrix} \begin{bmatrix} 2 \\ 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 1.364 \\ 2.182 \\ 1.000 \end{bmatrix}$$

Eigen Values and Eigen Vectors

An eigenvector of a linear transformation is a non-zero vector that changes at most by a scalar factor when that linear transformation is applied to it. The corresponding eigenvalue is the factor by which the eigenvector is scaled.

Eigen Values and Eigen Vectors

An eigenvector, corresponding to a real non-zero eigenvalue, points in a direction in which it is stretched by the transformation and the eigenvalue is the factor by which it is stretched.

Eigen Values and Eigen Vectors

Matrix

Eigen Value

$$\lambda \hat{\boldsymbol{e}} = A\hat{\boldsymbol{e}}$$

$$(A - \lambda I)\hat{\boldsymbol{e}} = \mathbf{0}$$

Eigen Vector

$$\|\hat{\boldsymbol{e}}\|_2 = 1$$

$$|A - \lambda I| = 0$$

Characteristic Equation

Eigen Values and Eigen Vectors

$$A = \begin{bmatrix} 3 & 5 \\ 7 & 9 \end{bmatrix}$$

$$\lambda_1, \lambda_2 = ?$$

$$\hat{\boldsymbol{e}}_1, \hat{\boldsymbol{e}}_2 = ?$$

Eigen Value Computation

$$A = \begin{bmatrix} 3 & 5 \\ 7 & 9 \end{bmatrix} \qquad |A - \lambda I| = 0$$

$$\begin{vmatrix} 3-\lambda & 5 \\ 7 & 9-\lambda \end{vmatrix} = 0$$

$$\lambda^2 - 12\lambda - 8 = 0 \longrightarrow \begin{pmatrix} 12.633 \\ -0.633 \end{pmatrix}$$

Characteristic Equation

Eigen Vector Computation: $\hat{\boldsymbol{e}}_{\mathbf{1}}$

$$\begin{bmatrix} 3 & 5 \\ 7 & 9 \end{bmatrix} \begin{bmatrix} e_{11} \\ e_{12} \end{bmatrix} = 12.633 \begin{bmatrix} e_{11} \\ e_{12} \end{bmatrix}$$

$$\lambda_1 = 12.633$$

$$3e_{11} + 5e_{12} = 12.633e_{11} \longrightarrow e_{12} = \frac{12.633 - 3}{5} e_{11} = 1.9266e_{11}$$

$$7e_{11} + 9e_{12} = 12.633e_{12} \longrightarrow e_{12} = \frac{7}{12.633 - 9} e_{11} = 1.9267e_{11}$$

Eigen Vector Computation: $\hat{\boldsymbol{e}}_{\mathbf{1}}$

$$\|\hat{\boldsymbol{e}}_1\|_2^2 = 1 \Rightarrow e_{11}^2 + e_{12}^2 = 1$$

$$e_{12} = 1.9266 e_{11}$$

$$e_{11}^2 + (1.9266 e_{11})^2 = 1 \longrightarrow \{1 + (1.9266)^2\} e_{11}^2 = 1$$

$$e_{11} = \frac{1}{\sqrt{1 + (1.9266)^2}} = 0.4607$$

$$e_{12} = 1.9266 \times 0.4607 = 0.8876$$

$$\hat{\boldsymbol{e}}_1 = \begin{bmatrix} 0.4607 \\ 0.8876 \end{bmatrix}$$

Eigen Vector Computation: $\hat{\boldsymbol{e}}_2$

$$\begin{bmatrix} 3 & 5 \\ 7 & 9 \end{bmatrix} \begin{bmatrix} e_{21} \\ e_{22} \end{bmatrix} = -0.633 \begin{bmatrix} e_{21} \\ e_{22} \end{bmatrix}$$

$$\lambda_2 = -0.633$$

$$3e_{21} + 5e_{22} = -0.633e_{21} \longrightarrow e_{22} = \frac{-0.633 - 3}{5} e_{21} = -0.7266e_{21}$$

$$7e_{21} + 9e_{22} = -0.633e_{22} \longrightarrow e_{22} = \frac{7}{-0.633 - 9} e_{21} = -0.7267e_{21}$$

Eigen Vector Computation: $\hat{\boldsymbol{e}}_{\mathbf{2}}$

$\|\hat{\boldsymbol{e}}_2\|_2^2 = 1 \Rightarrow e_{21}^2 + e_{22}^2 = 1$

$e_{22} = -0.7266 e_{21}$

$e_{21}^2 + (-0.7266 e_{21})^2 = 1 \longrightarrow \{1 + (0.7266)^2\} e_{21}^2 = 1$

$$e_{21} = \frac{1}{\sqrt{1 + (0.7266)^2}} = 0.809$$

$$\hat{\boldsymbol{e}}_2 = \begin{bmatrix} 0.809 \\ -0.5878 \end{bmatrix}$$

$$e_{22} = -0.7266 \times 0.809 = -0.5878$$

A Few Basic Definitions

Matrix A of size $n \times n$

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \Rightarrow$$

Eigen Vectors of A

$$\begin{matrix} \hat{\boldsymbol{e}}_1, & \ldots & \hat{\boldsymbol{e}}_i, & \ldots & \hat{\boldsymbol{e}}_n \\ \downarrow & & \downarrow & & \downarrow \\ \lambda_1, & \ldots & \lambda_i, & \ldots & \lambda_n \end{matrix}$$

Eigen Values of A

$$tr(A) = \sum_{i=1}^{n} \lambda_i = \sum_{i=1}^{n} a_{ii}$$

$$det(A) = |A| = \prod_{i=1}^{n} \lambda_i$$

Positive (Semi)Definite Matrix

Matrix A of size $n \times n$

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

Eigen Vectors of A

$\hat{\boldsymbol{e}}_1, \ldots \hat{\boldsymbol{e}}_i, \ldots \hat{\boldsymbol{e}}_n$

$\lambda_1, \ldots \lambda_i, \ldots \lambda_n$

Eigen Values of A

All Eigen Values Positive

Matrix A is Positive Definite

$$\boldsymbol{x}^T A \boldsymbol{x} > 0 \ \forall \boldsymbol{x} \ \|\boldsymbol{x}\|_2 \neq 0$$

All Eigen Values Non-Negative

Matrix A is Positive Semi-Definite

$$\boldsymbol{x}^T A \boldsymbol{x} \geq 0 \ \forall \boldsymbol{x} \ \|\boldsymbol{x}\|_2 \neq 0$$

Negative (Semi)Definite Matrix

Matrix A of size $n \times n$

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

Eigen Vectors of A

$\hat{\boldsymbol{e}}_1, \ldots \hat{\boldsymbol{e}}_i, \ldots \hat{\boldsymbol{e}}_n$

$\lambda_1, \ldots \lambda_i, \ldots \lambda_n$

Eigen Values of A

All Eigen Values Negative

Matrix A is Negative Definite

$$\boldsymbol{x}^T A \boldsymbol{x} < 0 \ \forall \boldsymbol{x} \ \|\boldsymbol{x}\|_2 \neq 0$$

All Eigen Values Non-Positive

Matrix A is Negative Semi-Definite

$$\boldsymbol{x}^T A \boldsymbol{x} \leq 0 \ \forall \boldsymbol{x} \ \|\boldsymbol{x}\|_2 \neq 0$$

Matrix Diagonalization

Matrix A of size $n \times n$

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}$$

Eigen Vectors of A

$$\hat{\boldsymbol{e}}_1, \dots \hat{\boldsymbol{e}}_i, \dots \hat{\boldsymbol{e}}_n$$

$$\lambda_1, \dots \lambda_i, \dots \lambda_n$$

Eigen Values of A

Matrix of Eigen Vectors

$$Q = [\hat{\boldsymbol{e}}_1 \quad \cdots \quad \hat{\boldsymbol{e}}_i \quad \cdots \quad \hat{\boldsymbol{e}}_n]$$

Diagonal Matrix of Eigen Values

$$\hat{A} = \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{bmatrix}$$

Matrix Diagonalization

Matrix A of size $n \times n$

Diagonal Matrix of Eigen Values

$$\hat{A} = \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{bmatrix}$$

Matrix of Eigen Vectors

$$Q = [\hat{\boldsymbol{e}}_1 \quad \cdots \quad \hat{\boldsymbol{e}}_i \quad \cdots \quad \hat{\boldsymbol{e}}_n]$$

$$A = Q\hat{A}\mathrm{Q}^{\mathrm{T}}$$

$$\Downarrow$$

$$A = \sum_{i=1}^{n} \lambda_i \hat{\boldsymbol{e}}_i \hat{\boldsymbol{e}}_i^T$$

$$A = A^T$$

$$\Downarrow$$

$$[Q^T = Q^{-1}] \wedge [|Q| = 1]$$

Matrix-Vector Multiplication

- We consider the Square Matrix **M**
  - Of Size $n \times n$
  - Having Eigen Values $\lambda_i; i = 1, \cdots n$
  - Having Eigen Vectors $\hat{\boldsymbol{e}}_{\boldsymbol{i}}$; $i = 1, \cdots n$
- **M** is Symmetric and Positive Semidefinite
- In this case, $\langle \hat{\boldsymbol{e}}_{\boldsymbol{i}}, \hat{\boldsymbol{e}}_{\boldsymbol{j}} \rangle = \delta_K[i-j]$ (Kronecker Delta Function)

$$\delta_K[i-j] = \begin{Bmatrix} 1 & ; IF\ i = j \\ 0 & ; Otherwise \end{Bmatrix}$$

We will Discuss the Matrix Vector Multiplication **Y = MX**

Matrix-Vector Multiplication (Contd.)

$$\boldsymbol{Z} = a_1\boldsymbol{V_1} + a_2\boldsymbol{V_2} + \cdots + a_n\boldsymbol{V_n}$$

Linearly Independent Component Vectors in a **n**-dimensional space

The Eigen Vectors ($\hat{e}_i$) are Orthogonal and Linearly Independent

$$\boldsymbol{X} = \alpha_1\boldsymbol{\hat{e}_1} + \alpha_2\boldsymbol{\hat{e}_2} + \cdots + \alpha_n\boldsymbol{\hat{e}_n} = \sum_{\boldsymbol{i=1}}^{\boldsymbol{n}} \alpha_i\boldsymbol{\hat{e}_i}$$

$$\boldsymbol{Y} = \beta_1\boldsymbol{\hat{e}_1} + \beta_2\boldsymbol{\hat{e}_2} + \cdots + \beta_n\boldsymbol{\hat{e}_n} = \sum_{\boldsymbol{j=1}}^{\boldsymbol{n}} \beta_j\boldsymbol{\hat{e}_j}$$

These Orthogonal Directions (Vectors) Form the Basis

Matrix (A) is a Linear Operator

$$\boldsymbol{A}\left(\sum_{k=1}^{n} a_k\boldsymbol{V}_k\right) = \sum_{k=1}^{n} a_k(\boldsymbol{A}\boldsymbol{V}_k)$$

Matrix-Vector Multiplication

$$\boldsymbol{Y} = \boldsymbol{MX} = \boldsymbol{M}\left(\sum_{i=1}^{n} \alpha_i \hat{\boldsymbol{e}}_i\right) = \sum_{i=1}^{n} \alpha_i (M\hat{\boldsymbol{e}}_i)$$

Application of Linearity Property of Matrix Operator **M**

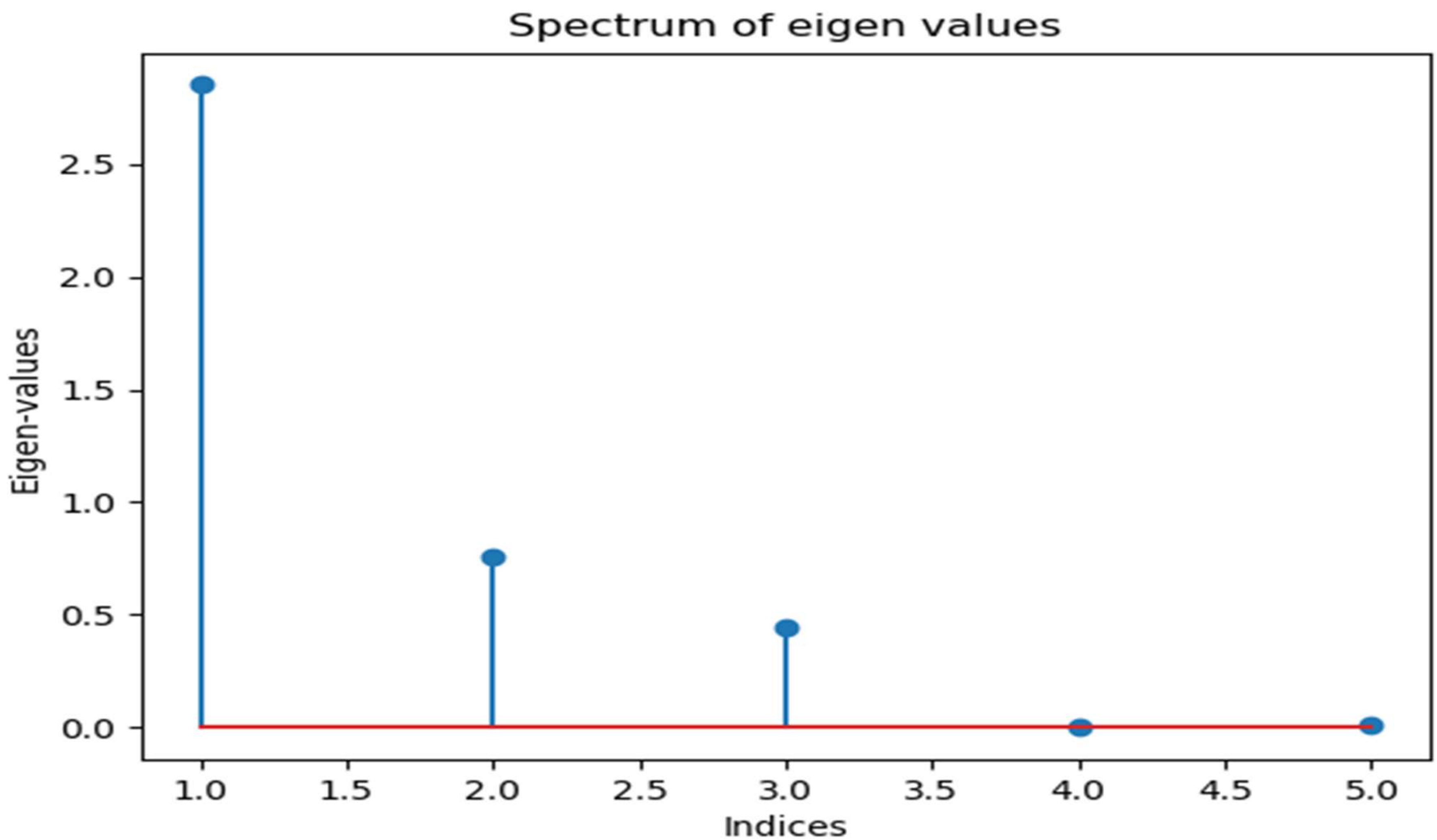
$$\boldsymbol{Y} = \sum_{j=1}^{n} \beta_j \hat{\boldsymbol{e}}_j = \sum_{i=1}^{n} \alpha_i (\boldsymbol{\lambda}_i \hat{\boldsymbol{e}}_i)$$

By Definition of Eigen Value and Eigen Vector of Matrix **M**

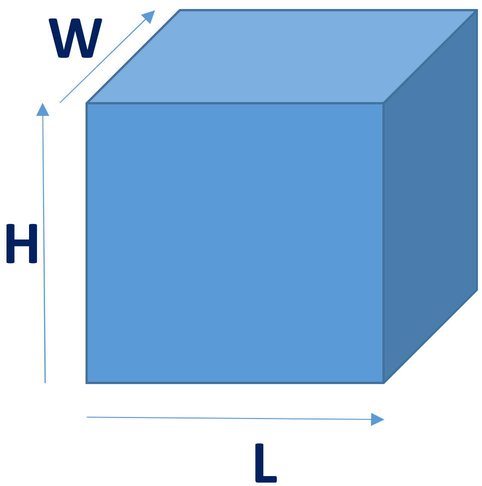
Inner Product with ($\hat{e}_r$) on Both Sides

$$\sum_{j=1}^{n} \beta_j \langle \hat{\boldsymbol{e}}_j, \hat{\boldsymbol{e}}_r \rangle = \sum_{i=1}^{n} \boldsymbol{\lambda}_i \alpha_i \langle \hat{\boldsymbol{e}}_i, \hat{\boldsymbol{e}}_r \rangle \Rightarrow \sum_{j=1}^{n} \beta_j \delta_K[j-r] = \sum_{i=1}^{n} \boldsymbol{\lambda}_i \alpha_i \delta_K[i-r]$$

$$\boxed{\beta_r = \lambda_r \alpha_r}$$


Spectrum of eigen values
Eigen-values
Indices
0.0
0.5
1.0
1.5
2.0
2.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0

Representing Space with Matrices



$$\mathbf{C}=\begin{bmatrix} L & 0 & 0 \\ 0 & W & 0 \\ 0 & 0 & H \end{bmatrix}$$

Eigen Values

L, W, H

Eigen Vectors

$$\begin{pmatrix}1\\0\\0\end{pmatrix}\begin{pmatrix}0\\1\\0\end{pmatrix}\begin{pmatrix}0\\0\\1\end{pmatrix}$$

Volume = $|C| = L \times W \times H$

Summary

- Introduction to Vector Space
- Norm and Metric Properties
- Matrix Operations
- Matrix as Operator, System and Space Descriptor
- Multiple Role of Eigen Values
- Eigen Vectors as Important Directions

Thank You